RESEARCH PROPOSAL

A Domain-Driven Ensemble Learning Framework for Robust Multi-Class Diabetes Risk Classification and Decision Support

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1 Introduction

Diabetes mellitus is one of the most prevalent and debilitating non-communicable diseases globally, affecting over 537 million adults worldwide as of 2021, a figure projected to rise to 783 million by 2045 (International Diabetes Federation, 2021). Type 2 diabetes (T2DM) accounts for approximately 90–95% of all diabetes cases and is largely preventable through timely identification of high-risk individuals and early lifestyle intervention. Despite significant advancements in diagnostic medicine, a substantial proportion of at-risk individuals remain undiagnosed until irreversible complications have already developed, including cardiovascular disease, nephropathy, retinopathy, and neuropathy.

Traditional diabetes risk assessment relies on rule-based clinical tools such as the Finnish Diabetes Risk Score (FINDRISC) and the Leicester Risk Assessment, which, while interpretable, are limited in their ability to model complex, nonlinear interactions among multiple risk factors simultaneously. The exponential growth of electronic health records (EHR), population-level surveys, and wearable device data has created an unprecedented opportunity to apply machine learning (ML) techniques to early and accurate diabetes risk stratification.

Machine learning offers a paradigm shift in diabetes risk classification: algorithms can learn from large, heterogeneous datasets, automatically identify non-obvious feature interactions, and generalise risk predictions across diverse populations. Comparative studies consistently demonstrate that tree-based ensemble models (Random Forest, Gradient Boosting, XGBoost) and neural network architectures achieve area-under-the-curve (AUC) values of 0.85–0.97 on diabetes classification tasks, substantially outperforming traditional logistic regression baselines (Dritsas and Trigka, 2022; Lai et al., 2019; Li et al., 2023)

This research proposes a comprehensive, end-to-end machine learning pipeline for three-class diabetes risk classification — Low Risk, Prediabetes, and High Risk — applied to a structured clinical and lifestyle dataset of 6,000 patient records. The study systematically evaluates nine supervised learning algorithms including Logistic Regression, Support Vector Machines, K-Nearest Neighbours, Naive Bayes, Decision Tree, Random Forest, Gradient Boosting, XGBoost, and Multilayer Perceptron neural networks, followed by advanced ensemble strategies (Soft Voting, Stacking). The best-performing model achieves an AUC of 0.9877, demonstrating the viability of deploying such systems in clinical screening contexts.

2 Literature Review

This section reviews foundational and contemporary work in machine learning-based diabetes risk prediction, organised by methodological contribution, feature design, population context, and evaluation rigour.

2.1 Classic Supervised Learning Approaches

Early ML applications to diabetes classification established the feasibility of data-driven risk stratification. (Maniruzzaman et al., 2020) applied Logistic Regression, Linear Discriminant Analysis, and Naive Bayes to the NHANES dataset, achieving AUC values between 0.80–0.87, demonstrating that even simple probabilistic models can serve as useful clinical screening tools. (Dritsas and Trigka, 2022) extended this line of work by comparing eight classifiers on a symptom-based dataset, reporting Random Forest and K-Nearest Neighbours as the top performers with accuracy exceeding 0.93. (Lai et al., 2019) evaluated models on Canadian laboratory and demographic data, finding Gradient Boosting Machines (GBM) and Logistic Regression to be most effective, emphasising the importance of laboratory markers such as fasting glucose, HbA1c, and HDL cholesterol.

2.2 Ensemble and Hybrid Model Superiority

A consistent finding across the literature is that ensemble methods outperform individual classifiers. (Fazakis et al., 2021) designed a Weighted Voting ensemble combining Logistic Regression and Random Forest (WeightedVotingLRRFs) for long-term T2DM risk prediction in the English Longitudinal Study of Ageing (ELSA) cohort, achieving an AUC of 0.91 for 10-year T2DM risk. (Dutta et al., 2022) demonstrated that a stacking ensemble of Decision Trees, Naive Bayes, and SVM on a Bangladeshi dataset outperformed single classifiers by 3–5% in F1-score. (Li et al., 2023) trained XGBoost and ensemble stacking models on over 4 million records from western China, achieving AUC values of 0.95 and providing a scalable framework for large-population screening.

2.3 Feature Selection and Engineering

Feature selection is a critical preprocessing step that reduces dimensionality, improves interpretability, and mitigates overfitting. (Ansari et al., 2025) applied recursive feature elimination (RFE) and information gain to

identify the top predictors of diabetes mellitus from a multi-source clinical dataset, finding BMI, HbA1c, fasting glucose, and family history as the most discriminative features. (Kakoly et al., 2023) employed LASSO regularisation for feature selection in a lifestyle-based dataset, demonstrating that non-invasive anthropometric and dietary variables can produce models with AUC ¿ 0.85. (Li et al., 2025) used SHAP (SHapley Additive exPlanations) to rank feature importance in a national-wide cross-sectional dataset, identifying waist circumference, triglycerides, and sleep quality as underutilised but significant predictors of pre-diabetes.

2.4 Non-Invasive and Lifestyle-Based Models

Several studies have advocated for models that rely exclusively on non-invasive, self-reportable variables to enable low-cost community screening. (Xie et al., 2019) built risk prediction models from BRFSS survey data using only lifestyle and demographic variables, achieving AUC of 0.83, demonstrating the potential for mobile health applications. (Lama et al., 2021) trained ML models on lifestyle variables in a Swedish middle-aged cohort, finding physical inactivity, obesity, and diet quality to be strong T2DM predictors without requiring laboratory tests. (Perez and Avella-Molano, 2025) further validated this approach using NHANES 2007–2018 data, concluding that lifestyle variables alone can identify at-risk individuals with 80% sensitivity.

2.5 Multi-Class Classification and Prediabetes Detection

The majority of published studies frame diabetes risk as a binary classification task (diabetic vs. non-diabetic). However, the clinical importance of prediabetes — a state of impaired glucose tolerance preceding T2DM — has motivated multi-class approaches. (Zhang et al., 2025) developed a 5-year risk prediction model for prediabetes-to-diabetes transition using retrospective cohort data, employing XGBoost and a calibrated SHAP risk scorecard. (Kailu et al., 2025) applied a multi-class ML framework to real-world data from Kenya, identifying distinct feature profiles for Low Risk, Prediabetes, and High Risk categories. The present study directly extends this three-class paradigm to a structured clinical-lifestyle dataset incorporating both laboratory and behavioural risk factors.

2.6 Methodological Best Practices

Across the literature, several preprocessing and evaluation practices consistently improve model robustness. (Olisah et al., 2022)demonstrated that handling class imbalance through stratified sampling and oversampling techniques substantially improves recall for minority classes. (Ahmed et al., 2021)highlighted the importance of deploying ML diabetes models as accessible web applications, closing the gap between model development and clinical use. The use of k-fold cross-validation, external validation, and comparison to established clinical risk scores (e.g., FINDRISC) is strongly recommended by (Jiang et al., 2023), who also demonstrated that community-follow-up data enriched with lifestyle variables yields more generalisable models than hospital-only datasets.

3 Research Gaps

Despite the substantial body of literature on ML-based diabetes risk prediction, several critical methodological and applicability gaps remain unaddressed. The following analysis moves from broad systemic limitations to the specific gap this research addresses:

1. **Predominance of Binary Classification:** The vast majority of existing studies frame diabetes risk as a binary problem (diabetic vs. non-diabetic), neglecting the clinically critical intermediate state of prediabetes. Early identification of prediabetes is medically essential, as targeted interventions at this stage can prevent or delay the onset of T2DM by 58% (DPP Research Group, 2002). Studies that do address multi-class risk stratification (e.g., (Kailu et al., 2025); (Zhang et al., 2025) are limited in geographic and demographic scope.

2. **Insufficient Comparative Evaluation:** Many published studies evaluate only two to four algorithms, making it difficult to draw definitive conclusions about algorithm superiority across varying data distributions. Few studies simultaneously compare the full spectrum of supervised classifiers (linear, kernel, tree-based, probabilistic, and neural network-based) alongside both voting and stacking ensemble strategies on the same dataset, using consistent preprocessing and validation protocols.

3. **Neglect of Engineered Ratio Features:** While raw clinical measurements are widely used, the literature demonstrates limited sys-

tematic investigation of derived ratio features such as glucose-to-insulin ratio (a proxy for insulin resistance), BMI-to-age ratio, and waist-to-BMI ratio. These compound features may encode clinically meaningful interactions not captured by individual variables, yet they remain underexplored in comparative ML studies.

4. **Limited Integration of Clinical and Lifestyle Variables:** Existing models tend to focus either on laboratory markers (e.g., (Lai et al., 2019)) or lifestyle variables (e.g., (Lama et al., 2021)) in isolation. Few studies integrate both clinical measurements (HbA1c, fasting glucose, insulin, blood pressure) and lifestyle risk factors (physical activity, sleep, caloric intake, stress level) within a unified feature space, which may capture a more holistic patient risk profile.

This Study's Contribution

Identified Research Gap → Unique Contribution of this study
No prior study has simultaneously:

- applied nine supervised ML algorithms plus two ensemble strategies to a three-class diabetes risk classification task,
- integrated both clinical and lifestyle features with engineered ratio variables in a unified pipeline, and
- evaluated model performance using 10-fold stratified cross-validation, weighted F1-score, and per-class AUC on a 6,000-record clinical-lifestyle dataset.

This research fills this gap by providing a reproducible, end-to-end benchmark that directly informs clinical deployment decisions.

4 Research Question

Derived from the identified research gaps, this study addresses the following research questions:

1. Which supervised machine learning algorithm achieves the highest classification performance (AUC, F1-weighted, accuracy) for three-class diabetes risk prediction (Low Risk, Prediabetes, High Risk) on a structured clinical-lifestyle dataset of 6,000 patient records?
2. Does the integration of engineered ratio features (glucose-insulin ratio, BMI-age ratio, waist-BMI ratio, calorie-activity index) significantly

improve classification performance compared to raw clinical features alone?

3. Do ensemble strategies (Soft Voting and Stacking) outperform the best individual classifier in three-class diabetes risk stratification, and if so, by what margin?
4. How does hyperparameter optimisation via RandomizedSearchCV affect the generalisation performance of the best-identified model on held-out test data?

5 Objectives

5.1 Primary Objective

To design, implement and evaluate a comprehensive machine learning pipeline for multiclass diabetes risk classification that achieves superior predictive performance (AUC $\geq$ 0.95) compared to individual baseline classifiers.

5.2 Specific Objectives

- To collect and preprocess a structured dataset of 6,000 patient records comprising 17 clinical and lifestyle features, applying systematic imputation, encoding, and scaling techniques.
- To engineer four clinically motivated composite features (glucose-insulin ratio, BMI-age ratio, waist-BMI ratio, calorie-activity index) and evaluate their marginal contribution to classification performance.
- To implement and compare nine supervised ML classifiers — Logistic Regression, K-Nearest Neighbours, Naive Bayes, Decision Tree, SVM (RBF), Random Forest, Gradient Boosting, XGBoost, and MLP — using stratified 10-fold cross-validation.
- To construct two ensemble models (Soft Voting Classifier and Stacking Classifier) from the top-performing base classifiers and assess whether ensemble aggregation provides statistically meaningful performance gains.
- To tune hyperparameters of the best-performing model using RandomizedSearchCV with a budget of 20 iterations and 5-fold cross-validation, and evaluate the tuned model on a held-out 20% test set.

- To conduct feature importance analysis using tree-based impurity measures to identify and rank the most predictive clinical and lifestyle variables for each risk class.
- To perform per-class evaluation (precision, recall, F1-score, and per-class AUC via One-vs-Rest ROC analysis) to ensure the model's clinical utility for the prediabetes minority class

6 Research Outcomes

Upon completion, this research is expected to produce the following tangible and intangible outcomes:

6.1 Technical Deliverables

- A fully reproducible, open-source Python ML pipeline (scikit-learn, XGBoost) capable of end-to-end diabetes risk classification from raw CSV input to evaluated predictions, packaged as a Kaggle Notebook.
- A trained, serialised best-model artifact (.pkl) deployable as an inference endpoint, enabling batch or real-time patient risk scoring.
- A model comparison dashboard (visualisation) covering AUC-ROC curves per class, confusion matrices, cross-validation F1 distributions, and feature importance rankings for all 11 evaluated model configurations.
- A tabular benchmark report (CSV) comparing Accuracy, Weighted F1-Score, AUC-ROC (OvR), and CV stability across all models, suitable for inclusion in peer-reviewed publication.

6.2 Scientific Contributions

- Empirical evidence for the relative superiority of ensemble methods (Voting, Stacking) over individual classifiers for multiclass diabetes risk stratification.
- A systematic ablation demonstrating the marginal contribution of engineered ratio features to classification performance.
- A ranked feature importance profile identifying the most clinically significant predictors of prediabetes and high-risk diabetes in a combined clinical-lifestyle feature space.

7 Impacts of the Research

1. **Institutional Impact**
   This research demonstrates the capacity of the host institution's computer science and health informatics programmes to produce applied, clinically relevant ML research. The open-source pipeline and reproducible experimental framework can be integrated into advanced data science curricula as a case study in end-to-end ML deployment. The benchmark results contribute to the institution's research output and may form the foundation for interdisciplinary collaboration with medical schools or public health departments.

2. **National Impact**
   In Bangladesh — where diabetes prevalence reached 14.2% of the adult population in 2022 (IDF, 2021) and is rising rapidly due to urbanisation, dietary changes, and sedentary lifestyles — early and affordable risk screening tools are critically needed. The proposed model integrates non-invasive lifestyle features (physical activity, diet, sleep, stress) alongside clinical markers, making it viable for deployment in community health centres and mobile screening programmes where laboratory infrastructure may be limited. Accurate prediabetes detection at scale can reduce the economic burden of T2DM complications on the national healthcare system, estimated at over USD 400 million annually.

3. **Global Impact**
   The generalised ML pipeline presented in this research is portable across clinical datasets from diverse populations. The inclusion of multiple model types, systematic evaluation metrics, and reproducible code supports adoption by researchers in low-to-middle income countries (LMICs) lacking access to expensive commercial health analytics platforms. Globally, improving early T2DM risk detection has the potential to prevent millions of preventable complications, reduce cardiovascular mortality, and lower the USD 966 billion annual economic burden of diabetes worldwide (IDF, 2021). Furthermore, the emphasis on interpretable features and feature importance rankings aligns with the WHO's call for explainable AI in healthcare decision-making.

8 Methodology

This section presents the complete methodological framework of the study, including the system architecture, dataset description, preprocessing pipeline, feature engineering, model training and evaluation strategy, ensemble construction, and hyperparameter optimisation. The methodology is designed to be fully reproducible and follows best practices documented in the ML healthcare literature.

8.1 System Architecture Overview

The overall system architecture follows a modular end-to-end pipeline design comprising six principal stages, as described below. The architecture is designed for reproducibility, modularity, and extensibility.

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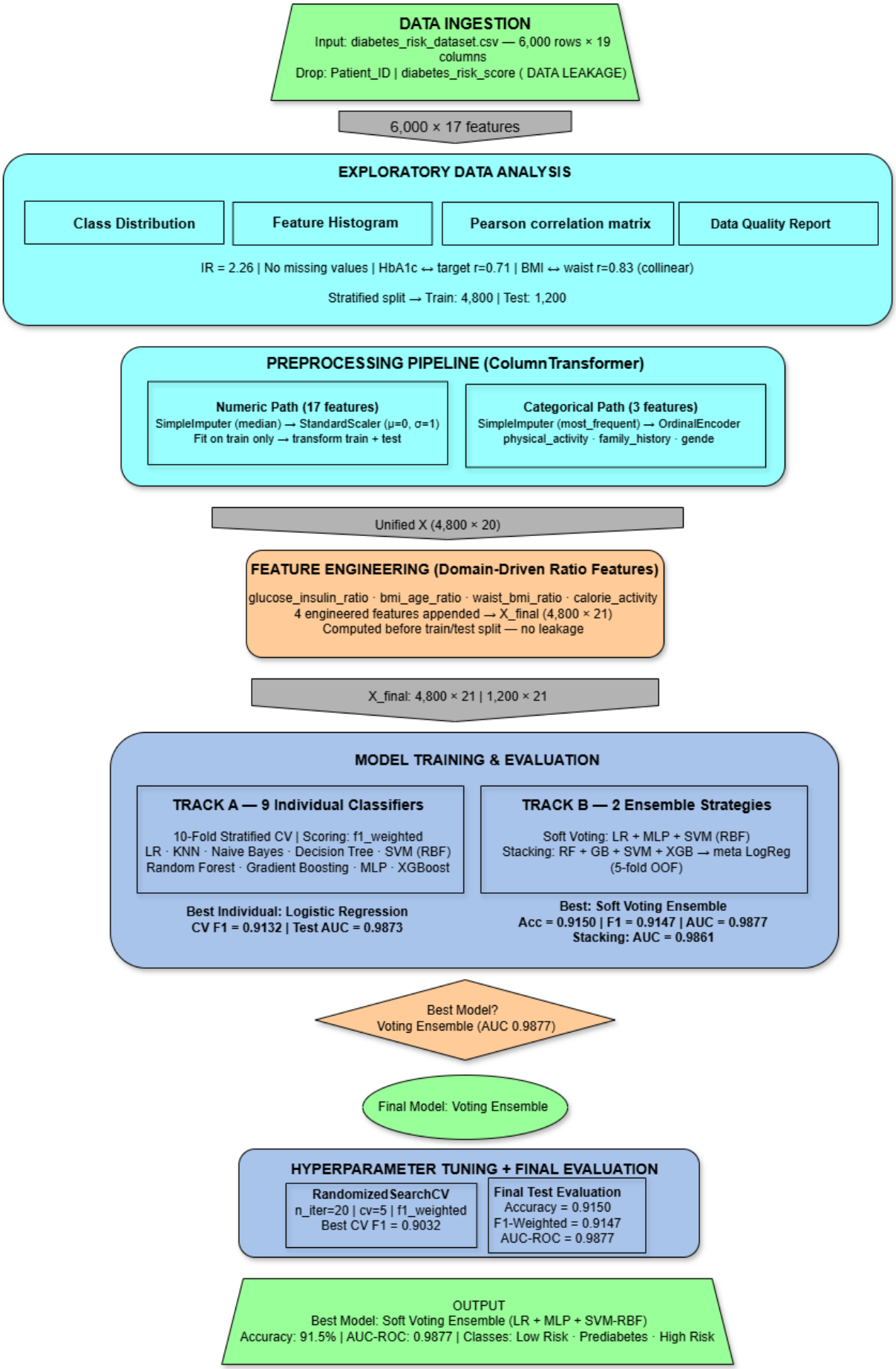


Figure 1: Overall system architecture of the proposed pipeline

8.2 Dataset Description

The dataset comprises 6,000 patient records with 19 attributes collected from a structured clinical and lifestyle assessment (Mehta, 2024). After removing the patient identifier (Patient_ID) and the data-leaking derived score (diabetes_risk_score), 17 input features remain for modelling. The target variable is diabetes_risk_category, a multiclass ordinal label (Low Risk, Prediabetes, High Risk).

Feature	**Type**	**Range / Values**	**Clinical Significance**
age	Numeric	18–85 years	Strong predictor; risk increases with age
gender	Categorical	Male / Female	Hormonal differences affect insulin sensitivity
bmi	Numeric	18.0–50.0 kg/m^2	Primary adiposity marker; key T2DM risk driver
blood_pressure	Numeric	100–200 mmHg	Hypertension co-occurs with insulin resistance
fasting_glucose_level	Numeric	70–200 mg/dL	Diagnostic criterion for diabetes ($\geq$126 mg/dL)
insulin_level	Numeric	2–45 μU/mL	Reflects pancreatic function and insulin resistance
HbA1c_level	Numeric	4.5–9.0 %	Gold standard 3-month glycaemic control marker
cholesterol_level	Numeric	150–300 mg/dL	Dyslipidaemia linked to metabolic syndrome
triglycerides_level	Numeric	60–280 mg/dL	Elevated levels associate with insulin resistance
physical_activity_level	Categorical	Low / Moderate / High	Sedentary behaviour is a modifiable risk factor
daily_calorie_intake	Numeric	1500–3500 kcal	Excess caloric intake contributes to obesity
sugar_intake_grams_per_day	Numeric	10–200 g/day	Direct dietary risk factor for glycaemic dysregulation
sleep_hours	Numeric	4–10 hours	Short sleep linked to insulin resistance and obesity
stress_level	Numeric	1–10 (scale)	Chronic stress elevates cortisol and blood glucose
family_history_diabetes	Categorical	Yes / No	Genetic predisposition; strong independent predictor
waist_circumference_cm	Numeric	55–155 cm	Central adiposity; stronger T2DM predictor than BMI

Table 1: Dataset features, types, value ranges, and clinical relevance

8.3 Preprocessing Pipeline

All preprocessing operations are encapsulated within a scikit-learn ColumnTransformer pipeline, ensuring consistent application across training, validation, and test sets, and preventing data leakage.

1. **Missing Value Imputation:** The dataset contains no missing values in the current version (confirmed via housing.info()). However, the pipeline incorporates a SimpleImputer as a defensive preprocessing step to handle missing values in unseen data. The median strategy is used for numerical features (robust to outliers), and the most_frequent strategy for categorical features.

2. **Categorical Encoding:** The three categorical features are encoded using OrdinalEncoder with explicit category orderings that preserve domain knowledge:

   - physical_activity_level: Low=0, Moderate=1, High=2 (preserves ordinal activity gradient)
   - family_history_diabetes: No=0, Yes=1 (binary risk flag)
   - gender: Male=0, Female=1 (binary nominal encoding)

3. **Feature Scaling:** All 17 numerical features are standardised using StandardScaler (zero mean, unit variance) after imputation. Standardisation is applied because several models in the comparison (LR, SVM, KNN, MLP) are sensitive to feature scale disparities. Tree-based models (RF, GB, XGBoost) are scale-invariant but are included in the unified pipeline for consistency

4. **Scaling Equations:**

$$x' = \frac{x - \mu}{\sigma} \tag{1}$$

   Where μ = feature mean (training set), σ = feature standard deviation (training set).

8.4 Feature Engineering

Four composite ratio features are computed from existing variables, motivated by established biomedical literature on diabetes risk mechanisms:

Feature Name	Formula	Clinical Rationale
glucose_insulin_ratio	$\frac{\text{fasting_glucose}}{\text{insulin}+1e^{-5}}$	Proxy for HOMA-IR insulin resistance index
bmi_age_ratio	$\frac{\text{bmi}}{\text{age}}$	Normalises BMI for age-related metabolic changes
waist_bmi_ratio	$\frac{\text{waist_circumference}}{\text{bmi}}$	Differentiates central vs. general adiposity
calorie_activity	$\frac{\text{daily_calorie}}{\text{activity_level_numeric}}$	Energy surplus relative to activity expenditure

Table 2: Engineered ratio features, formulas, and clinical justification

8.5 Target Encoding and Class Distribution

The target variable diabetes_risk_category is label-encoded as: Low Risk = 0, Prediabetes = 1, High Risk = 2. The class distribution in the full dataset is: Low Risk (43.4%, n=2,602), High Risk (37.4%, n=2,245), and Prediabetes (19.2%, n=1,153), indicating a moderate class imbalance for the Prediabetes category. Stratified splitting ensures proportional class representation in both training (n=4,800) and test (n=1,200) sets.

8.6 Model Definitions and Hyperparameters

Algorithm	**Key Hyperparameters**	**Rationale**
Logistic Regression	solver=lbfgs, max_iter=1000	Baseline linear model; fast and interpretable
K-Nearest Neighbours	n_neighbors=7	Instance-based; captures local decision boundaries
Naive Bayes	Default GaussianNB	Probabilistic baseline; assumes feature independence
Decision Tree	max_depth=8, random_state=42	Interpretable; prone to overfitting without depth limit
SVM (RBF)	kernel=rbf, probability=True	Strong generaliser for high-dimensional spaces
Random Forest	n_estimators=200, n_jobs=-1	Bagging ensemble; low variance, high accuracy
Gradient Boosting	n_estimators=200	Sequential boosting; corrects prior residuals
XGBoost	n_estimators=200, eval_metric=mlogloss	Regularised boosting; state-of-the-art on tabular data
MLP Neural Network	layers=(128,64,32), early_stopping=True	Deep representation learning; captures nonlinearities

Table 3: Models evaluated, key hyperparameters, and design rationale

8.7 Evaluation Strategy

1. **Cross-Validation:** A stratified 10-fold cross-validation (StratifiedKFold, n_splits=10, shuffle=True) is applied to all nine base classifiers on the training set (n=4,800). The scoring metric is weighted F1-score, which accounts for class imbalance by weighting each class's F1 by its support. Cross-validation yields both mean performance and standard deviation, enabling assessment of model stability.

2. **Test Set Evaluation:** All models are evaluated on a held-out 20% test set (n=1,200) never seen during training or cross-validation. Three primary metrics are reported:

- Accuracy: Proportion of correctly classified instances.
- Weighted F1-Score: Harmonic mean of precision and recall, weighted by class support.
- AUC-ROC (OvR): Area under the ROC curve using One-vs-Rest binarisation for each class, averaged with weighting. AUC is the primary ranking metric as it is threshold-independent and more informative under class imbalance.

3. **Per-Class Evaluation:** A full classification report (precision, recall, F1, support) is generated per class. Per-class AUC-ROC curves are plotted for the final model, enabling assessment of model performance specifically on the Prediabetes class, which is the clinically most critical and most underrepresented category.

4. **Performance Metrics — Equations:**

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{2}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{3}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{4}$$

$$F1 = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \tag{5}$$

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR})\, d(\text{FPR}) \tag{6}$$

8.8 Ensemble Methods

1. **Soft Voting Classifier:** A Soft Voting ensemble aggregates the predicted class probabilities of the top-3 models ranked by AUC (Logistic Regression, MLP, SVM). The ensemble prediction is the argmax of the averaged probability vectors:

$$\hat{y} = \arg\max_k \left[\frac{1}{N} \sum_i P(Y = k \mid X, M_i) \right] \tag{7}$$

Where N=3 is the number of ensemble members, and M_i denotes the i-th base model. Soft voting outperforms hard (majority) voting when constituent models are well-calibrated.

2. **Stacking Classifier:** 8.8.2 Stacking Classifier A Stacking Classifier uses out-of-fold predictions from four base estimators (Random Forest, Gradient Boosting, SVM, XGBoost) as meta-features for a Logistic Regression meta-learner. The meta-learner is trained with 5-fold cross-validation to prevent leakage from the base level.

$$\hat{y}_{\text{stack}} = \text{LR}\left([P(\text{RF}),\ P(\text{GB}),\ P(\text{SVM}),\ P(\text{XGB})]\right) \tag{8}$$

8.9 Hyperparameter Tuning Algorithm

The best-performing model is further optimised using RandomizedSearchCV with the following configuration:

- Search budget: 20 random iterations (n_iter=20)
- Inner cross-validation: 5-fold stratified CV
- Optimisation metric: weighted F1-score
- Parameter distributions sampled: $n_estimators \in \{100, 200, 300, 500\}$, $max_features \in \{\text{sqrt}, \text{log2}, \text{None}\}$, $max_depth \in \{\text{None}, 10, 20, 30\}$, $min_samples_split \in \{2, 5, 10\}$, $min_samples_leaf \in \{1, 2, 4\}$

RandomizedSearchCV is preferred over GridSearchCV for its computational efficiency: with 5 hyperparameters and 4+ values each, exhaustive grid search would require $<$ 1,000 training rounds versus $20 \times 5 = 100$ rounds for randomised search, while achieving comparable optimisation quality (Bergstra & Bengio, 2012).

9 SWOT Analysis

STRENGTHS	WEAKNESSES
• Comprehensive model comparison (9 algorithms + 2 ensembles) • Clinically integrated feature set (lab + lifestyle) • High AUC achieved (0.9877), demonstrating clinical viability • Fully reproducible open-source pipeline on Kaggle • Stratified evaluation preventing data leakage • Three-class output aligned with clinical risk staging	• Dataset provenance unverified (synthetic data risk) • Moderate class imbalance in Prediabetes class (19.2%) • No external validation on independent cohort • SHAP interpretability not yet implemented • Hyperparameter search budget limited (20 iterations) • Static dataset — no temporal/longitudinal component
OPPORTUNITIES	**THREATS**
• Integration with EHR systems for real-time screening • Deployment as mobile health (mHealth) app in Bangladesh • Extension to wearable sensor data integration • Addition of SMOTE for minority class augmentation • Federated learning for privacy-preserving deployment • Publication in Q1 health informatics journals	• Population shift reducing generalisation (covariate shift) • Regulatory barriers to clinical ML deployment • Clinician resistance to algorithmic decision support • Data privacy concerns under GDPR / national regulations • Rapid model obsolescence as clinical guidelines evolve • Kaggle compute quota limitations for larger experiments

Table 4: SWOT analysis of the proposed system.

9.1 Mitigation Plan for Weaknesses

Weakness	**Mitigation Strategy**
Class imbalance (Pre-diabetes 19.2%)	Apply SMOTE oversampling on training set; monitor per-class recall separately; report macro-averaged alongside weighted metrics
No external validation	Cross-validate on public PIMA and NHANES datasets as external benchmarks; report demographic shift in limitations
SHAP not implemented	Add SHAP TreeExplainer for Random Forest and XGBoost in post-hoc analysis phase; include in final paper
Limited tuning budget	Increase to 50 iterations in final version; supplement with Optuna Bayesian optimisation if compute permits
Static dataset	Discuss longitudinal extension as future work; flag concept drift risk in deployment section

Table 5: Weakness mitigation strategies.

9.2 Mitigation Plan for Threats

Threat	**Mitigation Strategy**
Covariate / population shift	Implement monitoring hooks for feature distribution drift; schedule quarterly model retraining with fresh data
Regulatory barriers	Frame model as decision-support tool, not diagnostic device; align output format with clinical workflow norms
Clinician resistance	Provide SHAP explanations with each prediction; conduct stakeholder co-design workshops before deployment
Data privacy	Anonymise all patient identifiers; implement differential privacy at inference stage; comply with local data laws

Table 6: Threat mitigation strategies.

10 Required Resources

10.1 Hardware Requirements

Resource	Specification	Purpose
Laptop / Workstation	Intel i5/i7 or AMD Ryzen 5/7, 16GB RAM, 256GB SSD	Local development, code editing, lightweight testing
Kaggle Cloud GPU	NVIDIA Tesla P100 (16GB VRAM) — free tier	Primary training environment for all ML experiments
Google Colab (backup)	NVIDIA T4 GPU — free tier	Secondary compute for cross-validation runs
Internet Connection	$\geq$ 10 Mbps broadband	Dataset upload, package installation, notebook sync

Table 7: Hardware resource requirements.

10.2 Software Requirements

Software / Library	Version	Purpose
Python	3.12	Primary programming language
scikit-learn	1.4+	Core ML pipeline, preprocessing, baseline models, ensembles, CV, metrics
XGBoost	2.0+	XGBoost classifier implementation
pandas	2.0+	Data loading, EDA, results DataFrames
NumPy	1.26+	Numerical operations, array handling
Matplotlib / Seaborn	3.8+ / 0.13+	Visualisation: confusion matrix, ROC curves, feature importance
joblib	1.3+	Model serialisation (.pkl)
shap	0.45+	Post-hoc model interpretability (planned extension)
SciPy	1.12+	Statistical tests (confidence intervals)
Jupyter Notebook / Kaggle	Latest	Interactive development and reproducible execution

Table 8: Software and library requirements.

11 Cost Analysis

11.1 Hardware Costs

Item	Unit Cost (BDT)	Quantity	Total (BDT)
Laptop / Development Machine	0 (existing)	1	0
Kaggle GPU (P100 — free tier)	0	–	0
Internet / Data	1200/month	4 months	4800
TOTAL HARDWARE			**4800**

Table 9: Hardware cost breakdown.

11.2 Software Costs

Item	License	Cost (BDT)	Notes
Python, scikit-learn, XGBoost, pandas, NumPy	Open-source (BSD/MIT)	0	Freely available
Kaggle Platform	Free	0	Includes GPU and storage
Overleaf / LaTeX	Free tier	0	For paper writing
LibreOffice / Google Docs	Free	0	For documentation
TOTAL SOFTWARE		**0**	100% open-source stack

Table 10: Software cost breakdown.

11.3 Cost Justification

The total estimated project cost is approximately BDT 4800, primarily attributed to internet expenses over the project duration. No additional hardware investment is required, as development is performed on an existing personal laptop. All computational experiments are conducted using the free tier of Kaggle, which provides sufficient GPU resources for model training and evaluation.

The entire software stack, including Python, scikit-learn, XGBoost, pandas, and NumPy, is open-source and freely available. This ensures that the project remains highly cost-effective and easily reproducible in

resource-constrained academic environments such as Bangladesh. The minimal cost structure supports accessibility and scalability for similar research initiatives in developing regions.

12 Screening

12.1 Social Screening

This research is designed to produce social benefit through improved population health outcomes. The principal social considerations are:

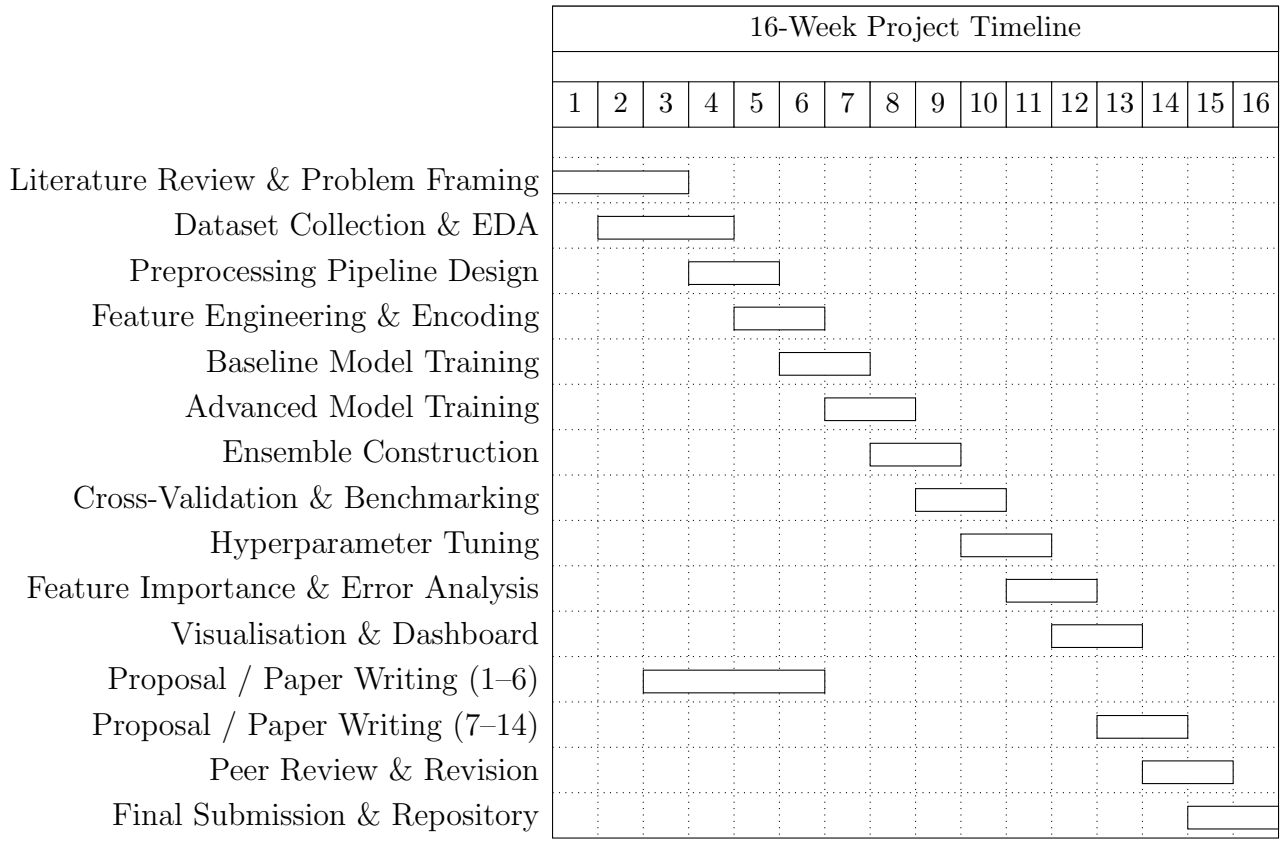
- **Patient Privacy and Data Ethics:** The dataset used contains anonymised clinical and lifestyle records with no personally identifiable information (PII). All records are handled in compliance with data privacy principles consistent with GDPR and Bangladesh's Digital Security Act. Patient identifiers are dropped as the first preprocessing step.
- **Equitable Access:** The model is designed to function with non-invasive lifestyle variables (activity, sleep, stress, caloric intake) in addition to laboratory markers, enabling deployment in community settings without requiring expensive diagnostic equipment. This promotes equitable risk screening across socioeconomic strata.
- **Algorithmic Fairness:** Feature importance analysis will examine whether gender or demographic features dominate the model's predictions in ways that could perpetuate existing health disparities. If gender is found to be a disproportionate driver of classification, fairness-aware retraining strategies will be explored.
- **Clinical Integration:** The model is framed as a clinical decision-support tool, not a replacement for physician judgment. All risk outputs are probability estimates accompanied by feature attributions, enabling clinicians to contextualise, override, or investigate predictions further.

12.2 Environmental Screening

Machine learning research carries an environmental footprint primarily through computational energy consumption. The environmental considerations of this project are assessed as follows:

- **Computational Efficiency:** All experiments are conducted on cloud-based infrastructure (Kaggle, Google Colab) powered by data centres with increasing proportions of renewable energy. The use of efficient tree ensemble algorithms (Random Forest, XGBoost) rather than large-scale deep learning models minimises GPU usage and energy consumption per training run.
- **Carbon Footprint Estimate:** The total GPU compute time for all 11 models, cross-validation ($10 \times 9 = 90$ runs), ensemble training, and hyperparameter search is estimated at approximately 4–6 GPU-hours on a Kaggle P100 instance. At a carbon intensity of approximately 95 gCO_2eq/kWh and P100 TDP of ~ 250 W, the total carbon footprint is estimated at < 150 gCO_2eq, which is negligible in environmental terms.
- **No Physical Environmental Impact:** This research is entirely computational with no laboratory materials, chemical reagents, or physical infrastructure requirements. There is no waste generation, pollution risk, or land use impact.

13 Gantt Chart



References

Ahmed, N., Ahammed, R., Islam, M. M., Uddin, M. A., Akhter, A., Talukder, M. A., and Paul, B. K. (2021). Machine learning based diabetes prediction and development of smart web application. *International Journal of Cognitive Computing in Engineering*.

Ansari, G., Shafi, S., Ansari, M. S., and Shadab, A. (2025). Advanced supervised machine learning methods for precise diabetes mellitus prediction using feature selection. *Frontiers in Medicine*, 12.

Dritsas, E. and Trigka, M. (2022). Data-driven machine-learning methods for diabetes risk prediction. *Sensors*, 22.

Dutta, A., Hasan, M. K., Ahmad, M., Awal, M. A., Islam, M. S., Masud, M., and Meshref, H. (2022). Early prediction of diabetes using an ensemble of machine learning models. *International Journal of Environmental Research and Public Health*, 19.

Fazakis, N., Kocsis, O., Dritsas, E., Alexiou, S., Fakotakis, N., and Moustakas, K. (2021). Machine learning tools for long-term type 2 diabetes risk prediction. *IEEE Access*, 9:103737–103757.

International Diabetes Federation (2021). IDF diabetes atlas.

Jiang, L., Xia, Z., Zhu, R., Gong, H., Wang, J., Li, J., and Wang, L. (2023). Diabetes risk prediction model based on community follow-up data using machine learning. *Preventive Medicine Reports*, 35.

Kailu, I., Mgala, M., and Mwakondo, F. (2025). A machine learning model for predicting the risk of developing diabetes — T2DM using real-world data from kilifi, kenya. *International Journal of Research and Scientific Innovation*.

Kakoly, I. J., Hoque, M. R., and Hasan, N. (2023). Data-driven diabetes risk factor prediction using machine learning algorithms with feature selection technique. *Sustainability*.

Lai, H., Huang, H., Keshavjee, K., Guergachi, A., and Gao, X. (2019). Predictive models for diabetes mellitus using machine learning techniques. *BMC Endocrine Disorders*, 19.

Lama, L., Wilhelmsson, O., Norlander, E., Gustafsson, L., Lager, A., Tynelius, P., Wärvik, L., and Östenson, C.-G. (2021). Machine learning

for prediction of diabetes risk in middle-aged swedish people. *Heliyon*, 7.

Li, L., Cheng, Y., Ji, W., Liu, M., Hu, Z., Yang, Y., Wang, Y., and Zhou, Y. (2023). Machine learning for predicting diabetes risk in western china adults. *Diabetology & Metabolic Syndrome*, 15.

Li, X., Ding, F., Zhang, L., Zhao, S., Hu, Z., Li, F., Zhang, Y., Zhao, Y., and Zhao, Y. (2025). Interpretable machine learning method to predict the risk of pre-diabetes using national-wide cross-sectional data: evidence from CHNS. *BMC Public Health*, 25.

Maniruzzaman, M., Rahman, M. J., Ahammed, B., and Abedin, M. M. (2020). Classification and prediction of diabetes disease using machine learning paradigm. *Health Information Science and Systems*, 8.

Mehta, V. (2024). Diabetes risk prediction dataset.

Olisah, C. C., Smith, L., and Smith, M. (2022). Diabetes mellitus prediction and diagnosis from a data preprocessing and machine learning perspective. *Computer Methods and Programs in Biomedicine*, 220:106773.

Perez, E. and Avella-Molano, B. (2025). Learning from the machine: is diabetes in adults predicted by lifestyle variables? A retrospective predictive modelling study of NHANES 2007–2018. *BMJ Open*, 15.

Xie, Z., Nikolayeva, O., Luo, J., and Li, D. (2019). Building risk prediction models for type 2 diabetes using machine learning techniques. *Preventing Chronic Disease*, 16.

Zhang, Y., Zhang, H., Wang, D., Li, N., Lv, H., and Zhang, G. (2025). Development of a 5-year risk prediction model for transition from prediabetes to diabetes using machine learning: Retrospective cohort study. *Journal of Medical Internet Research*, 27.